Exhaustive inconsistency: A and not A both refute Gamma

The Open Logic Project contributors

openlogic:first-order-logic/sequent-calculus:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-first-order-logic-sequent-calculus/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument:

Editorial note. The label prop:provability-exhaustive names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own prop environment in the unit source.

Allowed standard prerequisites (editorial).

- Right negation rule. From A, Gamma_0 sequent (empty) one derives Gamma_0 sequent (not A).. The displayed derivation applies Right-not to pi_0 before cutting against pi_1.
- Cut rule. From Gamma_0 sequent (not A) and (not A), Gamma_1 sequent (empty) one derives Gamma_0, Gamma_1 sequent (empty).. The cut composes the Right-not result with pi_1.

Proposition 0.1. *If* $\Gamma \cup \{A\}$ *and* $\Gamma \cup \{\neg A\}$ *are both inconsistent, then* Γ *is inconsistent.*

Proof. There are finite sets $\Gamma_0 \subseteq \Gamma$ and $\Gamma_1 \subseteq \Gamma$ and **LK**-derivations π_0 and π_1 of $A, \Gamma_0 \Rightarrow$ and $\neg A, \Gamma_1 \Rightarrow$, respectively. We can then derive

$$\dfrac{\dfrac{\begin{array}{c}\vdots\\ \pi_0\\ \vdots\\ A, \Gamma_0\end{array}}{\Gamma_0 \neg A} \qquad \begin{array}{c}\vdots\\ \pi_1\\ \vdots\\ \neg A, \Gamma_1\end{array}}{\Gamma_0, \Gamma_1}\ \text{Cut}$$

Since $\Gamma_0 \subseteq \Gamma$ and $\Gamma_1 \subseteq \Gamma$, $\Gamma_0 \cup \Gamma_1 \subseteq \Gamma$. Hence Γ is inconsistent. □